Ask or Intervene: A Benchmark for Mixed-Initiative Vision-and-Language Navigation

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Abstract

Vision-and-language navigation (VLN) typically assumes that a robot receives a fixed instruction before execution and then navigates autonomously. Real service robots instead operate under evolving human intent: a user may correct a mistaken turn, add a constraint, replace the destination, or answer a robot's request for assistance. Importantly, the user need not continuously monitor the robot, but must be able to intervene whenever necessary. We introduce MI-VLN, a benchmark for *mixed-initiative* VLN in indoor service environments. The benchmark models two complementary interaction directions: robot-initiated help requests and human-initiated interventions. It evaluates three tracks—human-initiated, robot-initiated, and fully mixed-initiative navigation—under asynchronous user availability. To isolate the value of interaction, each episode contains a shared pre-event trajectory and counterfactual post-event branches with matched initial conditions. We define intervention windows, dynamic task specifications, interaction-aware metrics, and a reproducible annotation protocol based on Habitat 3.0 and HSSD scenes. We further propose a baseline agent that combines visual-spatial memory, dynamic task memory, an initiative manager, and a language-conditioned navigator. Experiments are designed to measure navigation success, recovery efficiency, communication cost, intervention timing, and robustness to delayed human responses. [TBD: Insert the main quantitative finding after completing experiments.]

1 Introduction

An instruction such as "go to the kitchen and stop beside the table" is rarely immutable in a real home. While the robot is moving, a user may notice that the intended table was ambiguous, ask the robot to avoid a wet floor, or change the destination because the household task has changed. Conversely, the robot may recognize that its observation is ambiguous and ask whether it should take the left or right corridor. These interactions form a continuous decision process in which control of the dialogue can shift between the human and the robot.

Most VLN benchmarks evaluate a different setting: an agent receives a fixed natural-language instruction and is judged primarily by whether it reaches a static goal [1, 8, 6]. Interactive VLN work has studied dialogue-based assistance or robot help-seeking [14, 9, 15], while online robot correction has studied how language modifies plans or policies [12, 3]. However, a practical service robot must support both directions of initiative in the same episode. It must decide when to act, when to ask, how to incorporate unsolicited corrections, and how to proceed when a human is temporarily unavailable.

This paper introduces MI-VLN, a benchmark that treats interaction as part of the navigation problem rather than as an external reset mechanism. A human may intervene at any point, but is not assumed to watch every step. Likewise, the robot may request assistance, but unnecessary questions incur a cost. Messages update a structured task state without erasing the agent's navigation history. The benchmark therefore exposes a central trade-off: autonomous progress versus timely, useful communication.

Our contributions are:

- We formulate mixed-initiative VLN with bidirectional initiative, asynchronous human availability, and a dynamic language-defined task state.
- We design a household-service benchmark with three evaluation tracks, controlled intervention windows, counterfactual branches, and event-level annotations.

Episode timeline. The robot receives an initial instruction and navigates autonomously. At an event window, either (a) the human intervenes with a correction or a new constraint, or (b) the robot asks a targeted question and may receive a delayed answer. The interaction updates the current task specification; the physical state and navigation history are preserved. Evaluation continues until success, failure, timeout, or explicit termination.

Figure 1: Mixed-initiative navigation as a continuous, interruptible process. The benchmark does not require continuous human observation.

- We introduce metrics that jointly evaluate task success, recovery, communication cost, timing quality, and human effort.
- We provide a modular baseline and an experimental protocol for studying when an agent should ask, how it should adapt, and when human intervention is most valuable.

2 Related Work

Vision-and-language navigation. R2R established instruction-guided navigation in photo-realistic indoor environments [1]. Subsequent benchmarks expanded language, path diversity, and continuous control [8, 6, 7]. These settings usually treat the instruction and goal as fixed throughout an episode. MI-VLNinstead evaluates how an agent responds when intent changes during execution.

Dialogue and help-seeking in navigation. CVDN studies navigation through question–answer dialogue [14]. HANNA allows an agent to request help from an assistant [9], and later work studies help-request generation and collaborative navigation [15, 16, 13]. These works motivate the robot-initiated component of MI-VLN. Our benchmark adds unsolicited human-initiated intervention, temporary human unavailability, and direct comparison between initiative policies.

Online correction and shared autonomy. Language can update a robot policy or repair an erroneous plan during execution [3, 12, 4]. Shared-autonomy research further emphasizes adjustable control and human effort [2]. We translate these ideas to embodied VLN, where the agent must ground an evolving task specification in egocentric observations and long-horizon spatial memory.

Embodied simulation. Habitat provides a scalable platform for embodied AI [11]; Habitat 3.0 extends it toward collaborative human–robot tasks [10]. HSSD supplies high-quality, semantically annotated residential scenes [5]. We use these assets to construct reproducible household-service episodes.

3 Task Formulation

3.1 Interaction Model

At time t, the agent receives an egocentric observation

$$o_t = (I_t^{\text{rgb}}, I_t^{\text{depth}}, p_t), \tag{1}$$

where p_t denotes pose information when available. The interaction history is $H_t = \{u_0, m_1, \ldots, m_t\}$, where u_0 is the initial instruction and each m_i is an optional human or robot message.

The agent chooses from navigation and communication actions:

$$a_t \in \mathcal{A}_{\text{nav}} \cup \mathcal{A}_{\text{ask}}, \tag{2}$$

where $\mathcal{A}_{\text{nav}}$ contains movement and stop actions, and $\mathcal{A}_{\text{ask}}$ contains typed help requests. A human message may be a correction, route constraint, subgoal insertion, goal replacement, clarification, confirmation, or abort command.

Human availability is represented by $h_t \in \{0, 1\}$. If the agent asks when $h_t = 0$, the question may remain unanswered, be answered after a sampled delay, or expire according to the evaluation condition. This prevents the benchmark from equating "intervenable" with "continuously supervised."

3.2 Dynamic Task State

Instead of repeatedly concatenating all dialogue, we maintain a structured task state

$$B_t = (g_t, \mathcal{S}_t, \mathcal{C}_t, \mathcal{R}_t, q_t), \tag{3}$$

where g_t is the current goal, $\mathcal{S}_t$ is an ordered subgoal list, $\mathcal{C}_t$ contains route and safety constraints, $\mathcal{R}_t$ contains resolved references, and q_t is the current interaction status. A message induces an update

$$B_t = f_{\text{update}}(B_{t-1}, m_t). \tag{4}$$

Crucially, this update does not reset the physical state or erase visual and spatial memory.

3.3 Evaluation Tracks

1. **Human-Initiated (HI):** the robot cannot ask questions, but must process interventions arriving during navigation.
2. **Robot-Initiated (RI):** the task is fixed unless the robot asks for assistance; asking decisions and question utility are evaluated.
3. **Mixed-Initiative (MI):** both parties may initiate interaction, possibly under delayed or missing human responses.

   The agent seeks to maximize task utility

$$J = R_{\text{task}} - \lambda_{\text{path}} C_{\text{path}} - \lambda_{\text{comm}} C_{\text{comm}} - \lambda_{\text{risk}} C_{\text{risk}}, \tag{5}$$

rather than maximizing success at the cost of unlimited human assistance.

4 The MI-VLN Benchmark

4.1 Environment and Embodiment

The benchmark uses Habitat 3.0 with HSSD residential scenes. The primary embodiment is a single mobile service robot with RGB-D sensing and discrete short-horizon actions: move forward, turn left, turn right, look up, look down, and stop. This constrained action space makes intervention effects easier to replay and compare. A continuous-control extension is left for future work.

4.2 Base Household Tasks

Episodes cover navigation patterns common in domestic service:

- **Room and object approach:** reach a kitchen counter, bedside table, sofa, charging station, or storage shelf.
- **Person-oriented service:** navigate to a resident's location, wait nearby, or return to the user after visiting another room.
- **Multi-stage service:** visit a source location and then a delivery location, represented initially as ordered navigation subgoals.
- **Constraint-aware routing:** avoid a room, doorway, fragile area, occupied space, or temporarily blocked corridor.

Type	Example and state change
Directional correction	"Not that hallway; turn back and take the door on your left."
Reference repair	"By the table, I meant the small table beside the sofa."
Constraint addition	"Do not enter the nursery; the baby is sleeping."
Subgoal insertion	"Before the kitchen, stop at the charging dock."
Goal replacement	"Forget the bedroom; come back to the living room."
Priority or abort	"Stop there" or "Return immediately."

Table 1: Human-initiated intervention types in the initial benchmark.

- **Dynamic intent:** replace the destination, reorder subgoals, cancel an obsolete task, or redirect the robot to an urgent location.

The first release focuses on navigation and does not require physical object manipulation, although task descriptions may supply realistic service context.

4.3 Intervention Taxonomy

Each intervention is labeled by its semantic operation:

Robot questions are similarly typed as direction, landmark, reference, constraint, goal, or confirmation queries. Templates may be used for controlled training data, while test messages include human paraphrases.

4.4 Selecting Intervention Times

Interventions should be consequential but not arbitrary. We define candidate windows from observable episode events:

1. **Pre-error ambiguity:** before a junction with two plausible routes or referents.
2. **Recoverable deviation:** after the agent commits to an incorrect route but before recovery becomes trivial or impossible.
3. **Constraint relevance:** shortly before entering a newly prohibited region.
4. **Intent change:** after meaningful progress, when a replacement goal produces a nontrivial replanning problem.
5. **Safety-critical boundary:** before collision-prone or human-occupied areas, subject to conservative simulator rules.

For each window, we sample early, middle, and late intervention points. Let $\Delta C(t)$ denote the additional optimal path cost caused by waiting until t to intervene. Episodes are retained only when intervention changes the optimal continuation by at least threshold ϵ and still permits task completion. This yields meaningful timing comparisons without selecting times from a particular learned agent's failures alone.

4.5 Counterfactual Episode Branches

Each benchmark item stores a shared prefix state s_τ and several branches:

$$\mathcal{E} = \left(\xi_{0:\tau}, e_\tau, \xi_{\tau:T}^{\text{no-int}}, \xi_{\tau:T}^{\text{int}}, \xi_{\tau:T}^{\text{oracle}}\right), \tag{6}$$

where e_τ is an intervention opportunity. All branches begin from the same simulator state. This design supports paired evaluation of no interaction, actual interaction, and oracle-timed interaction, reducing variance caused by unrelated trajectory differences.

Split	Scenes	Base eps.	Branches	Utterances
Train	[TBD: N]	[TBD: N]	[TBD: N]	[TBD: N]
Val-Seen	[TBD: N]	[TBD: N]	[TBD: N]	[TBD: N]
Val-Unseen	[TBD: N]	[TBD: N]	[TBD: N]	[TBD: N]
Test-Unseen	[TBD: N]	[TBD: N]	[TBD: N]	[TBD: N]

Table 2: Planned benchmark statistics. Replace placeholders after generation and quality control.

4.6 Annotation and Quality Control

Annotation is performed through a Windows client connected to a Linux Habitat backend. Annotators receive a short observation clip, a top-down map available only to the annotation interface, the current instruction, and the intended state update. They produce a natural intervention or response and label its semantic operation. A second annotator verifies groundedness, executability, and consistency with the intended update. Paraphrase diversity is measured with lexical and semantic similarity statistics; contradictory or under-specified messages are adjudicated.

4.7 Data Splits

We partition by scene into train, validation-seen, validation-unseen, and test-unseen sets. Test splits additionally hold out selected combinations of intervention type, timing, and room transition to evaluate compositional generalization. No scene geometry or human-authored paraphrase is shared between train and unseen test sets.

4.8 Metrics

We report standard navigation success rate (SR), success weighted by path length (SPL), and navigation error. Interaction is measured by:

$$\text{RecoveryRate} = \frac{N_{\text{failed without interaction, succeeded with interaction}}}{N_{\text{recoverable episodes}}}, \tag{7}$$

$$\text{AskPrecision} = \frac{N_{\text{useful questions}}}{N_{\text{all questions}}}, \tag{8}$$

$$\text{AskRecall} = \frac{N_{\text{help needed and requested}}}{N_{\text{help needed}}}, \tag{9}$$

$$\text{UpdateConsistency} = \frac{N_{\text{actions consistent with latest task}}}{N_{\text{post-message actions}}}. \tag{10}$$

We also report question count, human intervention count, response delay, distance traveled after obsolete goals, constraint violations, and success gain per human message. A combined score is provided only as a secondary summary because its cost weights reflect deployment preferences.

5 A Mixed-Initiative Baseline

Our baseline contains four modules.

Visual-spatial memory. An RGB-D encoder produces observation features that are accumulated in a topological or metric memory M_t^{map}. This memory records visited locations, visual landmarks, and traversability evidence.

Dynamic task memory. A language encoder maps each new message to a semantic update over B_t. The update module predicts the affected fields and whether prior goals or constraints are superseded. The resulting memory preserves both the current task and an auditable change history.

Initiative manager. The manager estimates uncertainty U_t, expected failure risk F_t, and the expected value of a question V_t^{ask}. It asks only if

$$V_t^{\text{ask}} > C_{\text{ask}} + C_{\text{delay}}, \tag{11}$$

otherwise it navigates or waits. During human unavailability, it may continue with a conservative action, pause at a safe location, or postpone the question.

Navigator. A policy conditions on the current observation, spatial memory, task state, and interaction status:

$$\pi(a_t \mid o_t, M_t^{\text{map}}, B_t, H_t). \tag{12}$$

Training combines imitation learning on expert branches with reinforcement learning or offline preference optimization for communication cost. The total loss is

$$\mathcal{L} = \mathcal{L}_{\text{nav}} + \alpha\mathcal{L}_{\text{update}} + \beta\mathcal{L}_{\text{ask}} + \gamma\mathcal{L}_{\text{consistency}}. \tag{13}$$

6 Experiments

6.1 Research Questions

We organize experiments around five questions:

1. Does mid-execution language improve task success and recovery?
2. Can an agent identify when asking is more valuable than acting?
3. How do human- and robot-initiated interactions complement each other?
4. How sensitive are agents to intervention timing and delayed responses?
5. Does structured task memory improve adaptation over dialogue concatenation?

6.2 Baselines

- **No-Interaction VLN:** ignores all post-instruction messages.
- **Dialogue Concatenation:** appends messages to the original instruction and replans without explicit task-state updates.
- **Always Ask:** requests help at every candidate ambiguity.
- **Threshold Ask:** asks when navigation entropy exceeds a tuned threshold.
- **Oracle Initiative:** receives oracle intervention timing or answers, providing an upper bound.
- **MI-VLNBaseline:** uses dynamic task memory and the learned initiative manager described in Section 5.

6.3 Implementation Details

[TBD: Specify image encoder, language encoder, map representation, training schedule, hardware, random seeds, and all cost coefficients.] We will report mean and 95% confidence intervals over at least three seeds. Counterfactual branches will be evaluated with paired bootstrap tests. Model selection will use validation-unseen data without access to test intervention paraphrases.

Method	SR↑	SPL↑	Recovery↑	Viol.↓	Msgs.↓	Utility↑
No-Interaction VLN	TBD	TBD	TBD	TBD	0	TBD
Dialogue Concatenation	TBD	TBD	TBD	TBD	TBD	TBD
Always Ask	TBD	TBD	TBD	TBD	TBD	TBD
Threshold Ask	TBD	TBD	TBD	TBD	TBD	TBD
MI-VLNBaseline	TBD	TBD	TBD	TBD	TBD	TBD
Oracle Initiative	TBD	TBD	TBD	TBD	TBD	TBD

Table 3: Main results on the test-unseen mixed-initiative track. No empirical values are claimed in this draft.

Condition	SR	Recovery	Extra Dist.	Human Cost
HI only	TBD	TBD	TBD	TBD
RI only	TBD	TBD	TBD	TBD
MI	TBD	TBD	TBD	TBD
MI + delayed user	TBD	TBD	TBD	TBD
MI + intermittent user	TBD	TBD	TBD	TBD

Table 4: Planned comparison of initiative and availability conditions.

6.4 Main Results

6.5 Initiative Direction and Timing

We compare HI, RI, and MI tracks under matched episode prefixes. The analysis will distinguish failures caused by navigation, task-state updating, question selection, and unavailable users. Timing experiments evaluate early, middle, and late interventions within each event window. We hypothesize that earlier intervention reduces wasted motion, while a small amount of additional visual evidence can make some later interventions easier to ground; this hypothesis must be tested rather than assumed.

6.6 Ablations and Human Evaluation

We ablate structured task memory, interaction history, availability modeling, counterfactual training, and the communication-cost term. A human evaluation will measure whether questions are timely and answerable, whether interventions are correctly reflected in subsequent behavior, perceived workload, and trust. Annotators will view paired rollouts with randomized method identity.

7 Limitations and Broader Impact

Simulation cannot fully reproduce household noise, social expectations, sensor failure, or the consequences of unsafe motion. The initial benchmark uses short-horizon navigation actions and text interaction; speech recognition, gesture, manipulation, and multi-person authority are outside its first scope. Human messages may also encode cultural or household-specific assumptions. Models trained on synthetic or templated language risk learning annotation artifacts.

Mixed-initiative systems can reduce supervision burden, but poorly calibrated asking may annoy users, and incorrect compliance with an intervention may create safety risks. Benchmark success must therefore not be interpreted as certification for deployment. Real-world testing should use conservative motion limits, accessible stop controls, privacy-preserving data collection, and explicit rules for resolving conflicting instructions.

8 Conclusion

We presented MI-VLN, a benchmark proposal for household VLN in which either the robot or the human may initiate interaction. The formulation preserves continuous navigation state, supports asynchronous human availability, and represents language as updates to a dynamic task specification. The benchmark combines controlled intervention windows, counterfactual branches, and metrics that expose the trade-off between autonomy and human effort. The proposed baseline and experiments provide a concrete path toward evaluating robots that know when to ask and remain correctable whenever people choose to intervene.

A Illustrative Episode Record


```
{
  "episode_id": "hssd_0123",
  "scene_id": "scene_x",
  "initial_instruction": "Go to the table in the living room.",
  "event": {
    "type": "reference_repair",
    "window": [24, 31],
    "trigger_step": 27,
    "message": "I meant the small table beside the sofa."
  },
  "availability": {
    "mode": "intermittent",
    "response_delay_steps": 4
  },
  "task_update": {
    "replace_goal_reference": "small_table_near_sofa"
  },
  "branches": [
    "no_intervention",
    "human_intervention",
    "oracle_intervention"
  ]
}
```


B Additional Interaction Examples

Human-initiated correction. The robot approaches the right-hand corridor. The user says, "That route is blocked; use the corridor behind you." The correct behavior is to update the route constraint, preserve the destination, and replan from the current pose.

Robot-initiated clarification. At two visually similar doors, the robot asks, "Do you mean the bedroom beside the bathroom or the one at the end of the hall?" If no answer arrives, the agent must follow the configured conservative policy rather than assuming continuous supervision.

Mixed initiative. The robot asks about an ambiguous destination. Before answering that question, the user replaces the task: "Never mind; return to the living room." The system should invalidate the pending question and prioritize the new goal.

C Planned Release

The intended release includes episode-generation code, simulator configurations, state snapshots for counterfactual replay, train/validation/test metadata, annotation guidelines, validated utterances, baseline models, evaluation scripts, and model cards describing known limitations.

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